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by Ali Alzahmi - Friday, 15 August 2025, 5:23 PM

My initial post explained why Agent-Based Systems (ABS) help overcome the weaknesses of older centralised systems by empowering autonomous agent-based systems and proactive, reactive, and socially able agents (Gerges et al., 2021). ABS enables software agents to cooperate, bargain, and discover in dynamic, decentralised settings. So, it has application in multifaceted systems with heterogeneous participants and opposing goals, similar to smart grids, financial markets, urban mobility, and disaster response. I also emphasised the fact that the development of ABS with IoT, edge computing, and multi-agent simulations has more advanced capabilities that can offer resilient and scalable system structures, which are flexible and adaptable in the way they grow out of local interactions to global consequences in system architecture (Hayes & Grüter, 2023).

Fahad provided a comment on my apparent post as well as the relationship between theory and practice. He proposed touching on issues related to adoption, such as organisational challenges related to aligning agents with competing interests, and how it may integrate with future technologies such as predictive analytics or digital twins to turn ABS applications into more futuristic products. Ali Alhammadi concurred with my arguments and went further to say that ABSs in large-scale implementations could be subjected to communication bottlenecks, as well as autonomous agent conflicts (Ocker et al., 2021). He also suggested human-in-the-loop methods in vital fields, like healthcare and finance systems, so that decision-making can be explainable and accountable.

Koulthoum Hassan illuminated the need for preventive actions, including large-scale or casualty exercises, adaptive decision rules, secure communication, active surveillance, and regulation systems. She stated that such measures and ABS flexibility would be innovative and maintain operational reliability in sensitive areas. To sum it up, ABS is incredibly beneficial in decentralising and complex systems due to intelligent autonomy and bottom-up evolution (Pulikottil et al., 2023). Effective implementation needs proper control of the interplay between agents, communication, government, and reliable human control to guarantee dependability, safety, and responsibility.

References

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